

Engineering Reboot 2026

Week-02 Day-06 Field Manual v3.0
Conditionals

Mission Brief

Learn software decision-making using IF, IF-ELSE, and IF-ELIF-ELSE structures. Build a complete decision program and validate multiple execution paths.

Artifact Placement Matrix

high_temperature_alert.py -> Week-02/Exercises/Day-06/
temperature_status.py -> Week-02/Exercises/Day-06/
age_classification.py -> Week-02/Exercises/Day-06/
grade_evaluation.py -> Week-02/Exercises/Day-06/
decision_program.py -> Week-02/Exercises/Day-06/
Notes.md -> Week-02/Notes/Day-06/
Journal.md -> Week-02/Journal/Day-06/
Evidence -> Week-02/Day-06/Evidence/

Exercise 01 - High Temperature Alert

Filename: high_temperature_alert.py

Input Method: HARDCODED

Input: temperature = 85

Processing Rule:

If temperature > 80 then display:
Air conditioning recommended

Expected Console Output:

Air conditioning recommended

Evidence:

high_temperature_alert-Code.png

high_temperature_alert-ExpectedResult.png

Exercise 02 - Temperature Status Evaluation

Filename: temperature_status.py

Input: temperature = 65

If temperature > 80:
Air conditioning recommended

Else:

Temperature acceptable

Expected Output:

Temperature acceptable

Exercise 03 - Age Classification

Filename: age_classification.py

Validation Case A:

age = 16

Output: Minor

Validation Case B:

age = 21

Output: Adult

Evidence:

age_classification-Code.png

age_classification-Minor.png

age_classification-Adult.png

Exercise 04 - Grade Evaluation

Filename: grade_evaluation.py

Input: score = 83

90+ = Grade A

80-89 = Grade B

70-79 = Grade C

Below 70 = Needs Improvement

Expected Output:

Grade B

Exercise 05 - Decision Program

Filename: decision_program.py

Objective:

Evaluate voltage condition.

Validation Matrix:

100 -> Voltage Low

115 -> Voltage Normal

125 -> Voltage High

Evidence:

decision_program-Code.png

decision_program-Voltage-Low.png

decision_program-Voltage-Normal.png

decision_program-Voltage-High.png

Notes Template

Location:

Week-02/Notes/Day-06/Notes.md

Sections:
Concepts Learned
Key Commands
Key Observations
Lessons Learned
Questions For Future Study

Journal Template

Location:
Week-02/Journal/Day-06/Journal.md

Sections:
Summary
Challenges Encountered
Solutions Applied
Confidence Assessment
Reflection
Tomorrow Preparation

Git Workflow

```
git status
git add .
git commit -m "Day 06 conditionals completed"
git push
```

Verify GitHub repository.

Evidence:
Git-Commit.png
Git-Push-Success.png
GitHub-Remote-Verification.png

Expected Repository End-State

Week-02/
Exercises/Day-06/
high_temperature_alert.py
temperature_status.py
age_classification.py
grade_evaluation.py
decision_program.py

Notes/Day-06/Notes.md
Journal/Day-06/Journal.md
Day-06/Evidence/

Completion Checklist

- ☐ Exercise 01 complete
- ☐ Exercise 02 complete
- ☐ Exercise 03 complete
- ☐ Exercise 04 complete

- ☐ Exercise 05 complete
- ☐ Validation complete
- ☐ Notes complete
- ☐ Journal complete
- ☐ Evidence collected
- ☐ Git commit completed
- ☐ Git push completed
- ☐ GitHub verification completed